

Physics Origins 05: Toward the Schrödinger Equation

Part 2 — Forcing Out Planck's Constant from Black-Body Radiation

September 26, 2026

This is the prerequisite half of Chapter 05. It has one goal: to force h out, rather than importing it from somewhere else. The route is the classical black-body chain — count the modes, compute the mean energy per mode, find the clash with experiment, discard the assumption that energy is continuous, and use Wien's law to pin down the form of the energy quantum. Only after this does the h in de Broglie's postulate have a provenance, and only then can the main text of 05 begin.

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1 What this part has to settle

De Broglie's postulate $\lambda = h/p$ is the starting point of Chapter 05. Accepting it as given would leave an unexplained constant h sitting in the chain, and the derivation breaks there. So h has to be forced out first.

Deriving h is not the same as deriving de Broglie, and it is worth saying so before the work starts. Planck's route yields: the energy of an electromagnetic mode can only be $E = nh\nu$. That statement is about **light / the electromagnetic field**, and it relates **energy to frequency**. De Broglie needs $p = \hbar k$, which is about **massive particles** and relates **momentum to wavenumber**. Two gaps separate them:

1. energy–frequency $\rightarrow$ momentum–wavenumber: for photons this is bridged by $E = pc$ together with $\omega = ck$, and that step closes;
2. photon $\rightarrow$ electron: no derivation exists. This is de Broglie's analogical leap.

So the gain here is real but bounded: h acquires a provenance, and the outstanding IOUs drop from two (where h comes from; why particles are waves) to one. What remains is a genuine physical postulate.

The whole chain was walked backwards: Schrödinger $\leftarrow$ de Broglie $\leftarrow h \leftarrow$ black-body radiation $\leftarrow$ Maxwell's equations. From the Maxwell layer this part takes only two things: the three-dimensional wave equation (giving the dispersion relation $\omega = c|k|$) and the ideal-conductor boundary condition (giving the allowed values of k). Maxwell's equations themselves are a summary of experimental laws and are not derived here.

2 What a black body is

An object that only absorbs and never reflects, and which glows when heated.

"Black" refers to the absorbing side: every ray that lands on it is swallowed, nothing is reflected. Precisely because nothing is reflected, what you see coming off it is purely its own emission, independent of what was shone on it.

No perfect black body can be built, so a cavity with a small hole is used instead: light entering through the hole bounces around inside and essentially never escapes, which makes the hole a near-perfect absorbing surface. Conversely, heat the cavity to temperature T and what leaks out of the hole is black-body radiation. This is why the calculation ahead is about "which electromagnetic modes exist inside a cavity" — the apparatus itself is a cavity.

The key experimental fact: the spectrum depends only on temperature, not on what the cavity is made of or what shape it has. Red-hot iron and red-hot charcoal at the same temperature emit identically. It is precisely this material-independence that turns it into a purely theoretical problem — there are no material details to hide behind, so if the calculation disagrees it is the theory that is wrong.

Everyday instances: a glowing stove element, an incandescent bulb, the Sun (approximately), and a human body (at 37°C, emitting in the infrared). The higher the temperature, the shorter the wavelength of the peak — this is iron going from dull red to bright white.

2.1 Is a black hole a black body

Yes, and theoretically the most ideal kind.

The absorbing side is uncontroversial: nothing inside the horizon gets out, incoming light is absorbed with probability one, reflectivity is exactly zero. This beats any laboratory device. The emitting side is Hawking radiation (1974), whose spectrum is that of a black body at

$$T = \frac{\hbar c^3}{8\pi G M k_B}$$

Two caveats. First, the grey-body factor: the emitted particles must cross the gravitational potential barrier outside the horizon and some are scattered back, so the observed spectrum is the Planck spectrum times a frequency-dependent transmission coefficient. Second, this temperature is inversely proportional to mass; a solar-mass black hole sits at about 6×10^{-8} K, far below the CMB's 2.7 K, so it absorbs more than it radiates and grows rather than evaporates.

(The CMB, or cosmic microwave background: the universe became transparent at about 380,000 years, and the light released at that moment has travelled freely ever since, redshifted into microwaves. It is a 2.7 K black-body spectrum filling the whole sky.)

3 How it was measured

At the Physikalisch-Technische Reichsanstalt (PTR) in Berlin in the 1890s, by Lummer, Pringsheim, Rubens and Kurlbaum. The apparatus has three stages.

Source: an electrically heated cavity, blackened inside, with a small hole. Thermocouples measure the wall temperature, held stable above 1500 K — one of the engineering difficulties of the period.

Dispersion: the light from the hole passes through a prism or a diffraction grating and is spread out by frequency. Rotating the prism brings different frequencies onto the detector in turn, sweeping out the whole curve. The hard part is the infrared: glass prisms absorb it, so rock salt or fluoride crystals are needed. To reach further into the infrared, Rubens used the residual-ray method (Reststrahlen): letting light reflect repeatedly off a crystal surface, where a particular band has very high reflectivity and everything else is discarded, amounts to building a long-wavelength filter.

Detection: what is measured is the heating effect of the radiation, not light intensity. Two instruments dominate — the thermopile (a chain of thermocouples producing a voltage when heated) and the bolometer (a blackened metal strip whose resistance changes with heating, read out in a Wheatstone bridge). What the detector actually gives is a power; dividing by the bandwidth of the dispersing system, the detector area and the solid angle converts it into $\rho(\nu, T)$. This conversion and the associated calibration are the most laborious part of the experiment.

3.1 The mechanism of dispersion

Calling it “scattering” is inaccurate. Two mechanisms are used, neither of them scattering:

- **prism:** dispersive refraction. Different frequencies have different refractive indices in the medium, hence different exit angles.

- **grating:** diffraction and interference. The diffraction maximum of a given order falls at a different angle for each frequency.

Scattering (Rayleigh, Compton) is light being redirected by particles, with a random or distributed direction; it is not used for spectroscopy. The blue sky is scattering; the rainbow from a prism is not. The residual-ray method is a third mechanism again: strong reflection in a particular band (resonant with lattice vibrations), i.e. reflective filtering.

Decomposing light by frequency is the same operation as expanding a function in an orthogonal basis — the hardware version. A prism expands the incident light in the frequency basis, sending each component to a different exit angle. The machinery of Chapter 04 (“a function is a vector in an infinite-dimensional space, the orthogonal basis is $\sin(k_n x)$, the coefficients are obtained by projection”) makes its third appearance here.

3.2 One decisive afternoon

In October 1900 Rubens and Kurlbaum pushed the residual-ray measurements out to the far infrared (around $50\ \mu\text{m}$) and found a clear departure from Wien’s formula at long wavelengths, with the data tending instead towards proportionality to T . Rubens mentioned the result in passing while visiting Planck’s house that afternoon; Planck produced his interpolation formula that same evening and reported it to the society a few days later.

4 $\rho(\nu, T)$: the axes, and that zero

Horizontal axis: ν , frequency, in Hz (with $1\ \text{Hz} = 1/\text{s}$).

Vertical axis: ρ , energy density, in $\text{J}/(\text{m}^3 \cdot \text{Hz})$.

T is not an axis but a parameter — each temperature gives one curve, so the object is a family of curves. Experimentally one heats the cavity to a fixed temperature, measures a curve, then changes the temperature and measures another.

4.1 Why a single frequency carries zero energy

Because frequency is a continuous variable taking uncountably many values on any interval, while the total energy is finite. For infinitely many terms to sum to something finite, each term must be zero.

An analogy: a uniform string, 1 m long, 1 kg in mass. Asking “how much mass is at the point 0.3 m” gives zero — not because there is nothing there, but because a point has no length. What has mass is a *segment*: $[0.3, 0.31]$ carries 0.01 kg. So the linear density $\lambda = 1\ \text{kg}/\text{m}$ is meaningful and “the mass of a point” is not.

The spectrum works the same way: $\rho(\nu, T)$ is a density (energy per unit frequency), and what is meaningful is $\rho(\nu, T) d\nu$, the energy in the small band $[\nu, \nu + d\nu]$. The quantity playing the role of “length” in the analogy is the **frequency interval** (bandwidth) $d\nu$, in Hz.

One refinement: in a cavity of finite size the modes are in fact discrete, $\nu_n = (c/2L)n$. So one really can point at a single mode and speak of its energy, and that number is not zero. The zero belongs to the continuum limit $L \rightarrow \infty$; experimentally the modes are packed too densely to resolve, which is why the continuous description is used.

It follows that solar power, or any measured power, is an integral over a band of frequencies rather than the value at a single frequency. But the direction of causation matters: it is not that mathematics decreed the zero, it is that the spectrum is physically continuous, whence single points have measure zero and a density description becomes necessary. For a finite cavity the modes are discrete, “the energy of this one mode” is a nonzero number, and the density description is not needed at all.

5 The curve and three experimental facts

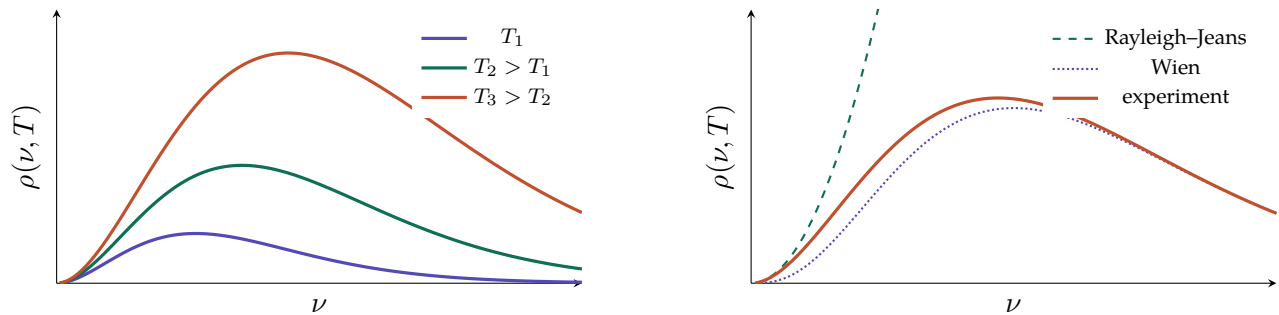


Figure 1 Left: a family of curves; raising the temperature lifts the whole curve and moves the peak to higher frequency. Right: three lines at one temperature. Rayleigh-Jeans matches at low frequency and diverges at high; Wien matches at high frequency and falls short at low; the experimental curve has a peak and a finite area.

At fixed T the curve rises from zero, peaks at some ν_{peak} , then falls rapidly back to zero. Three experimental facts are needed:

- **Wien’s displacement law** (1893): $\nu_{\text{peak}} \propto T$. This is red-hot going to white-hot.
- **Stefan–Boltzmann law** (1879 experimental, 1884 theoretical): the area under the curve $\propto T^4$.
- **The two asymptotics**: at short wavelengths Wien’s formula $\rho \propto \nu^3 e^{-a\nu/T}$ fits well; at long wavelengths Rubens measured, in October 1900, $\rho \propto T$, the first power of temperature — which Wien’s formula cannot produce.

The two dashed lines have different standing, and it is worth keeping them apart.

Wien’s formula was not derived from theory; Wien assembled it semi-empirically in 1896. It matched at high frequency so well that it was taken to be the final answer, until the far-infrared measurements of October 1900 showed the low-frequency end was wrong.

Rayleigh–Jeans is the genuine classical prediction: number of modes $\times k_B T$ per mode. It matches at low frequency and diverges at high — the ultraviolet catastrophe. Chronologically it is later than Planck: Rayleigh gave it in June 1900 and Jeans corrected the coefficient in 1905, so strictly speaking Planck was not “repairing Rayleigh–Jeans” when he assembled his formula.

Planck’s situation in October 1900 was therefore: an empirical formula that was very accurate at high frequency, plus fresh news that the low-frequency end behaves as $\rho \propto T$, and the two disagreed. What he needed was an interpolation correct at both ends. Not a fit to scattered points — a functional form inferred from two limiting behaviours, with the data points serving as the final check.

5.1 Why the low-frequency failure is invisible on a linear plot

At low frequency both curves are close to zero in absolute terms, negligible next to the peak, so the discrepancy is squashed out of sight. The ratio, however, diverges:

$$\frac{\rho_{\text{Planck}}}{\rho_{\text{Wien}}} = \frac{e^{\nu/T}}{e^{\nu/T} - 1} \rightarrow \frac{T}{\nu}, \quad \nu \rightarrow 0$$

The lower the frequency, the more badly Wien's formula undershoots — by factors of tens — but all of that happens in a region where the absolute value is tiny. This is exactly why the discovery waited until 1900: reaching that region required instruments that could work far into the infrared with sufficient sensitivity.

6 The mode count $g(\nu)$

6.1 What a mode is

A mode is a standing wave, that is, one permitted pattern of vibration.

Take the string of Chapter 04 as the analogy. On a string fixed at both ends the possible standing waves are not arbitrary: only the sequence $k_n = n\pi/L$ survives. Each n is one pattern, one mode. Electromagnetic waves in a cavity are the same thing in three dimensions — the walls are conducting, so the electric field must vanish on them, which plays the role of “fixed at both ends”, and only a particular set of standing waves fits inside.

Two things must be kept apart. The list of standing waves is “which waveforms can exist”, produced by the boundary condition, and the list is infinite. Decomposing an arbitrary waveform into a superposition of these is a separate matter (Chapter 04, section 5). Counting modes means counting the list.

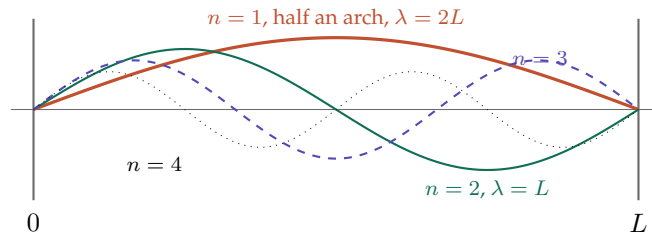


Figure 2 The first four modes on a string fixed at both ends. Every mode vanishes at both ends; n counts the half-waves fitted onto the string, $\lambda_n = 2L/n$ and $k_n = 2\pi/\lambda_n = n\pi/L$.

In one dimension each k is one mode and k corresponds one-to-one with ν . In three dimensions it does not: a standing wave has the form

$$\sin \frac{n_x \pi x}{L} \sin \frac{n_y \pi y}{L} \sin \frac{n_z \pi z}{L}, \quad \nu \propto \sqrt{n_x^2 + n_y^2 + n_z^2}$$

so (1, 2, 3), (2, 1, 3) and (3, 2, 1) — spatially quite different patterns — all give the same ν . That is where “many standing waves near one frequency” comes from.

6.2 What a mode means physically: an independent degree of freedom

The electromagnetic field in the cavity is not one undifferentiated lump. It decomposes into a set of mutually independent vibration patterns, each oscillating on its own without disturbing the others. The “count” counts those patterns.

This is the same thing as a degree of freedom (Chapter 02: “a degree of freedom is a component of the system that can vary independently”). One mode is one degree of freedom. So “there are $g(\nu)d\nu$ modes near frequency ν ” says: in this band, how many independently variable vibrational degrees of freedom the cavity has. Energy is distributed among degrees of freedom, so counting modes is counting the places energy can be put.

Several standing waves do of course superpose and interfere in space, but the **energy** does not mix: when the total energy is expanded over modes, the cross terms integrate to zero by orthogonality, and the total equals the sum of the individual mode energies. This is the precise content of “independent degree of freedom”, and it relies on the orthogonality established in Chapter 04.

6.3 The boundary condition

[IOU] The inner wall is taken to be a perfect conductor, with boundary condition $E_{\parallel} = 0$; the result is independent of material and shape, on the authority of Kirchhoff’s law (1859). Not descended into.

No electric field can exist inside a conductor (free electrons would move until it was cancelled), so the tangential component of the field must vanish right at the wall, which is what forces the nodes of the standing waves. The analogy with the string holds only in mathematical form: a string is clamped so its displacement vanishes, a cavity wall conducts so the field vanishes. Different mechanisms, identical boundary condition.

A perfect conductor does not transmit: an incident wave is totally reflected (the field decays exponentially inside the conductor). Real metals have small transmission and absorption losses, but these only give the modes a slight linewidth; they change neither the mode count nor the energy distribution. And the absorption and re-emission by the walls is precisely the mechanism of thermal equilibrium — with no energy exchange at all, equilibrium could never be reached.

6.4 The three factors

[IOU] The three-dimensional wave equation $\partial^2 \vec{E} / \partial t^2 = c^2 \nabla^2 \vec{E}$ is taken as given; it comes from Maxwell’s equations, not derived here. Only two consequences are used: after separation of variables each direction reduces to the one-dimensional case of Chapter 04, giving $k_i = n_i \pi / L$; and substituting a plane wave gives the dispersion relation $\omega = c|k|$, i.e. $\nu = c|k|/2\pi$.

One: the shell count $4\pi n^2 dn$. Draw each mode as a point with coordinates (n_x, n_y, n_z) , all positive integers; these points form a lattice of unit spacing. With $|k| = n\pi/L$ where $n = \sqrt{n_x^2 + n_y^2 + n_z^2}$ is the distance from the origin, the frequency depends on that distance alone. So “modes with frequency in $[\nu, \nu + d\nu]$ ” means “lattice points at distance in $[n, n + dn]$ ”, a spherical shell. Each lattice point occupies one unit cube, so the count $\approx$ the volume $= 4\pi n^2 dn$.

This is an approximation. Exact lattice-point counting in a ball is a number-theoretic problem with no elementary closed form; replacing count by volume incurs an error of the order of the surface area (relative error $\sim 1/n$), and for a cavity much larger than the wavelength n is astronomically

large. The rigorous version is Weyl's law: the leading term of the eigenvalue count depends only on the volume, with shape entering only at next order (proportional to surface area).

Two: only one octant. Since $\sin(-n_x\pi x/L) = -\sin(n_x\pi x/L)$ differs only by an overall sign, it is the same vibration pattern, so n_i and $-n_i$ must not be counted twice. Each index runs over positive integers, only the first octant of lattice points is valid, and that is $1/8$ of the shell. This is the three-dimensional version of “ $-n$ and $+n$ are the same mode” from Chapter 04, applied once per direction.

Three: two polarisations. For a given k , the electric field can point along two independent directions in the plane perpendicular to k . This follows from $\nabla \cdot \vec{E} = 0$, which gives $\vec{k} \cdot \vec{E} = 0$: the field must be transverse. The perpendicular plane is two-dimensional, so with two orthogonal basis directions every field direction is a combination of them. Hence one k carries two modes.

6.5 Putting them together

$$g(\nu)d\nu = 4\pi n^2 dn \cdot \frac{1}{8} \cdot 2, \quad n = \frac{2L\nu}{c}, \quad dn = \frac{2L}{c}d\nu$$

Substituting,

$$\pi \left(\frac{2L\nu}{c} \right)^2 \cdot \frac{2L}{c} d\nu = \frac{8\pi L^3 \nu^2}{c^3} d\nu$$

and dividing by the cavity volume L^3 gives the number of modes per unit volume per unit frequency:

$$\boxed{g(\nu) = \frac{8\pi\nu^2}{c^3}} \quad (1)$$

L cancels completely, as it must.

6.6 Two corollaries

The energy density does not depend on cavity size. The mode count is proportional to volume and the division by volume cancels it, so $\rho(\nu, T)$ depends only on ν and T . This is Kirchhoff's statement showing up concretely.

Nor on shape. A long box and a cube of equal volume carry the same number of modes. The intuition that “a long box fits more standing waves” holds only along one direction: the spacing of allowed k along the long edge is finer, giving more modes, but the other two directions are squeezed, with coarser spacing and fewer modes. In k -space each lattice point occupies

$$\frac{\pi}{L_x} \cdot \frac{\pi}{L_y} \cdot \frac{\pi}{L_z} = \frac{\pi^3}{V}$$

which depends on V alone.

7 The mean energy per mode

7.1 What $k_B T$ says

The cavity holds a collection of standing waves which continually exchange energy with the walls until thermal equilibrium is reached. The classical equipartition theorem gives: each mode receives, on average, $k_B T$, **independently of frequency**.

k_B is Boltzmann's constant (1.38×10^{-23} J/K), converting temperature into energy; $k_B T$ is “the portion of energy corresponding to temperature T ”, measured in joules, with no volume attached. This is the mean total energy of **one mode** — a mode fills the whole cavity and has no “local” part. Volume enters only later: summing the energies of all modes in a frequency band and dividing by volume and $d\nu$ produces ρ .

The “average” here is a time average: the mode exchanges energy with the wall continuously (the atoms in the wall both absorb and emit), so its instantaneous energy fluctuates. The fluctuation is objective, not a measurement error. Equilibrium does not mean nothing moves; it means the statistics do not change.

The energy itself is the energy of this standing wave's vibration — an oscillating electromagnetic field carries energy (electric plus magnetic), and the temperature sets how large it oscillates on average.

7.2 States and phase space

Mathematically a mode is a harmonic oscillator (Chapter 04: $\ddot{c}_n = -\omega^2 c_n$). Its state is given by two numbers:

$$E(x, p) = \frac{p^2}{2m} + \frac{1}{2}m\omega^2 x^2$$

where $p^2/2m$ is the kinetic term ($\frac{1}{2}mv^2$ rewritten in momentum) and $\frac{1}{2}m\omega^2 x^2$ the potential term. For a spring, x is the present displacement from equilibrium and p the present momentum; for an electromagnetic standing wave, x is the present field amplitude and p its rate of change. Note that x is the **instantaneous** displacement, not the maximum: the maximum is the amplitude A , and x ranges over $\pm A$.

[IOU] The m in $E = \frac{1}{2}m\omega^2 A^2$ is replaced, for an electromagnetic mode, by a constant of the field; it is not a mass and merely occupies the same slot. This does not affect the conclusion.

Classically x and p take continuous values, so E can be any real number from 0 to ∞ . That is the assumption Planck will later change.

7.3 Where the distribution comes from

Boltzmann distribution: the probability of a state of energy E is $\propto e^{-E/k_B T}$.

It follows from maximum entropy: maximise the entropy subject to the single constraint that the mean energy is fixed, apply Lagrange multipliers, and the solution is $P \propto e^{-\beta E}$ with the multiplier $\beta = 1/k_B T$ (derivation in Appendix A). The correspondence here is: the feature is $f = E$ (the energy itself, one-dimensional, given), the parameter is $\theta = \beta$, and the constraint is “mean energy = a fixed value”.

Two things must not be conflated. The “mean” in the constraint is the first moment of a distribution, not an average counted off one snapshot; and $f = E$ is the energy of a **single state**, whereas

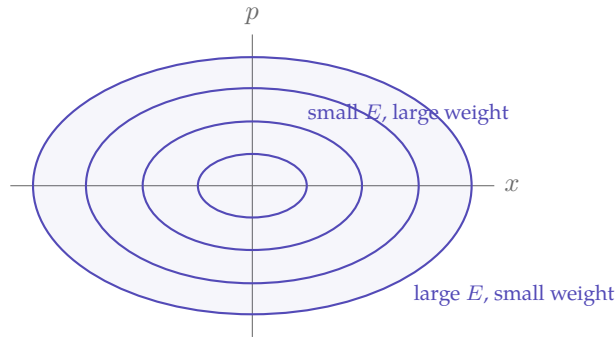


Figure 3 Phase space (x, p) : the curves of constant energy are ellipses, and the Boltzmann weight $e^{-\beta E}$ decays exponentially outwards. Classically the integral runs over the whole plane; Planck's modification amounts to keeping only the ellipses at $E = n\varepsilon$ and deleting everything in between.

the mean energy $\bar{E}$ is the right-hand side of the constraint.

Under the premise “only the mean energy is known, no further assumptions”, the solution is unique — maximum entropy is a strictly convex problem. Getting a different distribution requires changing the premise. Of course maximum entropy is a mathematical conclusion and whether nature obeys it is for experiment to decide; the empirical support for the Boltzmann distribution is strong (molecular speed distributions, atmospheric density with height, specific heats).

Historically Boltzmann himself derived it by microstate counting, well before Jaynes's maximum-entropy route of 1957; the two roads give the same distribution.

7.4 The standing of $k_B T$

[IOU] Classical equipartition gives $\bar{E} = k_B T$; the source is the expectation of energy under the Boltzmann distribution with continuous energy. That integral is not carried out here.

What does need stating is its relation to the constraint. The constraint direction is: given $\bar{E}$, solve for β . The reverse direction is: given β (that is, given the temperature), compute $\bar{E}$. The answer, $\bar{E} = 1/\beta = k_B T$, makes the two directions inverse functions of each other. But the statement “ $k_B T$ is that fixed value” is only available after the computation — without it one knows only that $\bar{E}$ and β correspond somehow, not that the correspondence is exactly $\bar{E} = 1/\beta$.

Moreover the result depends on the form of E : a free particle has only a kinetic term and gives $\bar{E} = \frac{1}{2}k_B T$; here there are two quadratic terms and the answer is $k_B T$. So $k_B T$ comes from “the energy is a sum of two quadratic terms”, not from maximum entropy as such.

It is precisely this correspondence that Planck changes.

8 Rayleigh–Jeans and the ultraviolet catastrophe

Multiplying the two pieces:

$$\rho(\nu, T) = g(\nu) \cdot \bar{E} = \frac{8\pi\nu^2}{c^3} k_B T \quad (2)$$

What the classical theory predicts experimentally: a curve rising monotonically with frequency, with no peak and infinite total energy. Opening the cavity should release an infinite amount of high-frequency radiation.

What is observed: a spectrum with a peak, falling rapidly to zero beyond it, with finite total energy. Working backwards, the mean energy of high-frequency modes must be far below $k_B T$, and increasingly so with frequency.

So the direct experimental signature of the conflict is: **whether the curve has a peak**. Classically it does not; experimentally it does.

The essential point is not that there are infinitely many degrees of freedom. The number of degrees of freedom grows as ν^2 , and each receives the same $k_B T$, so $\int \nu^2 d\nu$ diverges at high frequency. Were the high-frequency degrees of freedom to receive sufficiently little, there could still be infinitely many of them with a finite total.

9 Planck's modification

9.1 The assumption

[IOU] $E_n = n\varepsilon$ is an assumption, not derivable; Planck himself did not derive it.

The only change: the energy of a mode is no longer continuous but restricted to

$$E_n = n\varepsilon, \quad n = 0, 1, 2, 3, \dots$$

with ε the smallest unit of energy for that mode (one “portion”), left unspecified for now. The distribution is unchanged — the Boltzmann distribution is used as before — with a sum replacing the integral, which is the only structural difference. In the phase-space picture, the continuous ellipses become a discrete family, with only those at $E = n\varepsilon$ surviving. The maximum-entropy derivation itself is untouched: constraint, multiplier and exponential form all carry over; only the outcome space changes from continuous to countable.

9.2 Recomputing the mean energy

The partition function:

$$Z = \sum_{n=0}^{\infty} e^{-\beta n\varepsilon} = \sum_{n=0}^{\infty} \left(e^{-\beta\varepsilon} \right)^n = \frac{1}{1 - e^{-\beta\varepsilon}}$$

since $\beta\varepsilon > 0$ makes the ratio $r = e^{-\beta\varepsilon} \in (0, 1)$ and the geometric series converges. This is the entire benefit of discretisation: the continuous case requires Gaussian integrals, the discrete case is a geometric series.

Using $\bar{E} = -\partial(\log Z)/\partial\beta$:

$$\log Z = -\log(1 - e^{-\beta\varepsilon}), \quad \frac{\partial \log Z}{\partial \beta} = -\frac{\varepsilon e^{-\beta\varepsilon}}{1 - e^{-\beta\varepsilon}}$$

so that

$$\boxed{\bar{E} = \frac{\varepsilon}{e^{\beta\varepsilon} - 1} = \frac{\varepsilon}{e^{\varepsilon/k_B T} - 1}} \quad (3)$$

Only the combination $\varepsilon/k_B T$ appears.

9.3 The two limits

$\varepsilon \rightarrow 0$: using $e^x - 1 \approx x$ gives $\bar{E} \approx k_B T$. When the steps are too fine to see, equipartition is recovered — which is why the low-frequency end agrees with experiment.

$\varepsilon \gg k_B T$: the denominator is approximately $e^{\varepsilon/k_B T}$, giving $\bar{E} \approx \varepsilon e^{-\varepsilon/k_B T}$. The decay is exponential, far below $k_B T$, and faster than any polynomial growth.

The physical reading: the thermal bath can only hand this mode a whole portion ε at a time. When ε greatly exceeds $k_B T$, thermal fluctuations cannot assemble even one portion, the mode sits at $n = 0$ nearly all the time, and its mean energy tends to zero. In the continuous case there is no such threshold, so every mode collects its $k_B T$.

10 Forcing out $\varepsilon = h\nu$

10.1 Wien's law in its general form

Wien derived in 1893, from thermodynamics rather than by fitting:

$$\rho(\nu, T) = \nu^3 f(\nu/T) \quad (4)$$

with f an unknown function of a single variable.

[IOU] The derivation of this form (adiabatic compression plus the Doppler effect) is not descended into here. The idea: compressing a mirrored cavity full of radiation raises each mode's frequency as $1/L$, while the work done on the radiation raises the temperature as $1/L$ as well; the two scale together, so ν/T is an adiabatic invariant. Since ν/T does not change, the shape of the spectrum can depend on temperature only through that combination.

It supplies only a skeleton; it cannot determine f . So it explains how the curve changes with temperature (the peak moves right, the shape is self-similar) but not what the curve is. Determining f is exactly what Planck did.

This must be kept apart from its namesake: in 1896 Wien also assembled the specific formula $\rho \propto \nu^3 e^{-a\nu/T}$, one guess at f , right at high frequency and wrong at low. Both are called Wien's law.

10.2 The displacement law is a corollary

Holding T fixed and differentiating with respect to ν :

$$\frac{\partial \rho}{\partial \nu} = 3\nu^2 f\left(\frac{\nu}{T}\right) + \nu^3 f'\left(\frac{\nu}{T}\right) \cdot \frac{1}{T}$$

Setting this to zero and cancelling ν^2 (note that $\nu^3 \cdot \frac{1}{T}$ divided by ν^2 gives $\frac{\nu}{T}$, the same combination again):

$$3f(x) + xf'(x) = 0, \quad x = \nu/T$$

This equation contains neither T nor ν , only x . Its solution is a definite number x^* depending only on the shape of f . Hence

$$\nu_{\text{peak}} = x^* \cdot T$$

Change the temperature and the equation is unchanged, so the solution is unchanged and the peak moves in proportion to T . Everything turns on cancelling that ν^2 : afterwards ν and T appear only through ν/T , and the equation can no longer see their individual values.

10.3 Pinning down ε

Multiplying the two pieces:

$$\rho(\nu, T) = \frac{8\pi\nu^2}{c^3} \cdot \frac{\varepsilon}{e^{\varepsilon/k_B T} - 1}$$

This must be expressible as $\nu^3 f(\nu/T)$. The prefactor ν^2 already supplies two powers, so the remaining factor must supply the third and depend on nothing but ν/T :

- the numerator $\varepsilon(\nu)$ must be proportional to ν , or that third power never appears;
- writing $\varepsilon = h\nu$, the exponent becomes $h\nu/k_B T$, a function of ν/T , so the denominator complies automatically.

A counterexample: $\varepsilon = h\nu^2$ puts ν^2/T in the exponent, which is not a function of ν/T , and Wien's form breaks. Hence

$$\boxed{\varepsilon = h\nu} \tag{5}$$

The dimension of h follows from energy divided by frequency, i.e. J·s. We now have $E = nh\nu$.

Physically ε is the smallest step in a mode's energy: the mode can hold $0, \varepsilon, 2\varepsilon, \dots$ and nothing in between. This was later understood as the energy of one photon, with n the number of photons in that mode — all photons in a mode sharing the same energy, direction and polarisation, with arbitrarily many allowed in one mode (the signature of bosons). Still, reading n as “which energy step” fits the mathematics better than picturing a swarm of little balls.

10.4 “Modes carry energy” versus “a single point carries none”

Modes live on the integer lattice (n_x, n_y, n_z) and are countable; the frequency axis is a continuous real line. Each mode corresponds to a point on that axis, but the overwhelming majority of points correspond to no mode at all. So: the points that do carry modes carry energy, in integer multiples of $h\nu$; in the continuum approximation these points are smeared into a density $g(\nu)$, and only then does “the energy at a single point” become zero.

The two statements belong to different levels of description — one is about a single mode before smearing, the other about a density afterwards. As $L \rightarrow \infty$ the points become denser (the energy at each falls as $1/L^3$) while the product stays fixed.

One distinction to keep: the cavity size governs whether the **frequency spectrum** is discrete or continuous, and the spectrum becomes continuous as $L \rightarrow \infty$; whereas the discreteness of **energy** ($E = nh\nu$) is governed by h , has nothing to do with L , and never goes away.

(A note: this “two descriptions, discrete and continuous” language is modern. Nobody in 1900 proceeded this way — the spectrum was treated as continuous for integration, the modes as discrete for counting, the two used side by side without conflict and without any felt need to reconcile them.)

11 Planck’s formula

$$\rho(\nu, T) = \frac{8\pi h \nu^3}{c^3} \cdot \frac{1}{e^{h\nu/k_B T} - 1} \quad (6)$$

Checking against Wien’s form:

$$\rho = \nu^3 \cdot \frac{8\pi h}{c^3} \cdot \underbrace{\frac{1}{e^{(h/k_B)(\nu/T)} - 1}}_{f(\nu/T)}$$

which indeed involves only ν/T .

The two ends:

limit	result	corresponds to
$h\nu \ll k_B T$	$\rho \approx \frac{8\pi \nu^2}{c^3} k_B T$	Rayleigh–Jeans; matches the low-frequency data
$h\nu \gg k_B T$	$\rho \approx \frac{8\pi h \nu^3}{c^3} e^{-h\nu/k_B T}$	Wien’s formula, with $a = h/k_B$

The constant a that Wien assembled in 1896 finally acquires a provenance here.

11.1 The value of h

Only h and k_B remain unknown, and both follow from fitting the measured curve: the position of the peak fixes one combination, the height another. Planck obtained $h \approx 6.55 \times 10^{-34}$ J·s in 1900, within 1% of today’s $6.62607015 \times 10^{-34}$. The k_B he obtained at the same time yielded Avogadro’s number to better precision than any other method then available, which was among the strongest early evidence for the formula.

Since the 2019 SI revision, h is fixed by definition as an exact number and is used in turn to define the kilogram. Before that it was measured, chiefly via the Kibble balance and the silicon-sphere Avogadro route.

The standing of this step: h is not derived from any deeper principle, it is a fitted parameter. What can be derived is only that it **must exist** — Wien’s $\nu^3 f(\nu/T)$ form plus the assumption of discrete energy forces the existence of a constant with dimensions J·s.

Its meaning is a conversion rate between energy and frequency, just as c converts between space and time and k_B between temperature and energy. It tells you how big one portion is. If energy in nature were continuous, h would be zero — $h \neq 0$ is itself equivalent to the statement that the world is quantum.

A convenient order-of-magnitude check: $h\nu/k_B T = 1$ at room temperature (300 K) corresponds to about 6×10^{12} Hz, in the infrared; visible light is around 5×10^{14} Hz, giving $h\nu/k_B T \approx 80$ — which is why objects at room temperature do not glow visibly.

12 The historical route Planck actually took

The derivation above — pinning $\varepsilon = h\nu$ with Wien's form — is the textbook reconstruction: shorter and logically tighter, but not the historical path. What actually happened:

Step one (October 1900): assemble the formula. He worked with the second derivative of entropy with respect to energy, $\partial^2 S / \partial U^2$. Wien's formula corresponds to this quantity going as $1/U$; the low-frequency data ($\rho \propto T$) correspond to $1/U^2$. He added the two: $\partial^2 S / \partial U^2 = -a/[U(b+U)]$, integrated twice, and arrived at Planck's formula. The whole operation was at the level of the entropy function — no ε , no discretisation.

Step two (December 1900): look for an explanation. With the formula already correct, he asked what assumption could produce it. He used Boltzmann's combinatorial counting: divide the total energy into P portions of size ε , distribute them among N oscillators, count the arrangements, take the logarithm with Stirling's approximation to get the entropy. Requiring that entropy to agree with step one forces $\varepsilon \propto \nu$ — $\varepsilon \propto \nu$ **was forced out by the formula he had already assembled.**

So the real order is: experiment $\rightarrow$ assemble a formula $\rightarrow$ infer an assumption. $\varepsilon = h\nu$ only appears at the third step. The mathematical simplicity (the geometric series) is a later editorial tidying; the combinatorial counting he actually used is considerably more laborious.

As methodology, the pattern is: find a disagreement between observation and theory, go back and inspect the assumptions in the theory, change one of them, see whether the result closes mathematically, and then ask whether the changed assumption is true of the world. Planck did not first identify "energy is continuous" as the suspect assumption and then modify it; he assembled a formula first and inferred the assumption afterwards. Modifying the assumption was the outcome, not the starting point.

12.1 How it was received

Very few people noticed the tension, and those who did treated it not as a contradiction but as a strange assumption.

Planck himself was the most uneasy. For roughly a decade he tried to get rid of ε , hoping to show that discreteness was a mathematical device; in 1911 he put forward a "second theory" in which absorption was continuous and only emission quantised, to confine the quantum hypothesis as narrowly as possible. His colleagues largely let it pass: the formula was widely accepted (it fitted, and it yielded accurate values of k_B and Avogadro's number as a bonus), while the discreteness assumption was regarded as a dubious step in a derivation that most people chose not to interrogate.

Einstein was the exception. In 1905 he did not defend Planck but pointed out that Planck's derivation was internally inconsistent: Planck placed the quantisation on the **oscillators** (wall oscillators with energies $n h \nu$) while the electromagnetic field itself remained classical and continuous; yet the exchange of energy between oscillator and field was computed with classical electrodynamics, which permits arbitrarily small exchanges. The two cannot both hold.

Einstein's move was to relocate the quantisation from the oscillators to the **radiation field itself**. Leaving oscillators aside, he examined the entropy of radiation: in the Wien regime, the way that entropy varies with volume has exactly the form it would have for n independent particles of an ideal gas, with $n = E/(h\nu)$. The conclusion is that radiation itself consists of portions of energy $h\nu$. Planck's internal inconsistency then disappears.

The price is a conflict with the wave nature of light. Interference and diffraction show unambiguously that light is a wave, whereas “a swarm of independent energy quanta” is a particle picture. Einstein was entirely clear about this and called it a “heuristic viewpoint”. So the character of this step is: exchange a technical inconsistency for a deeper contradiction (wave–particle duality), while gaining a testable new prediction.

In 1907 he applied the same assumption to the specific heat of solids, carrying the quantum hypothesis into a domain with no light in it. What changed the situation decisively was the first Solvay Conference of 1911, devoted to the quantum problem, followed by Bohr’s atomic model in 1913.

12.2 How discreteness was seen experimentally

Classical experiments cannot see it because the steps are too fine: for a macroscopic oscillator $h\nu$ is a dozen orders of magnitude below $k_B T$, and the levels are indistinguishable from a continuum. So “classical physics shows energy is continuous” really means “no steps are measurable at that scale”. Three direct verifications:

- **The photoelectric effect** (predicted 1905, verified precisely by Millikan in 1916): the maximum kinetic energy of the ejected electrons depends on the frequency of the light and not on its intensity, and below a threshold frequency no electrons emerge however intense the light. The continuous picture predicts that enough intensity for long enough would accumulate the required energy; experiment refutes it. The slope of the line $E_{\max}$ versus ν is h — a second, independent route to its value.
- **Specific heat of solids** (Einstein 1907, Debye 1912): classical equipartition predicts a constant specific heat (Dulong–Petit), but measurements show it tending to zero at low temperature. The cause is the same mechanism: when $k_B T$ falls below $h\nu$ the high-frequency vibrational modes freeze out. This verifies energy discreteness in **mechanical** vibration, with no light involved.
- **Atomic spectra**: gases emit only at particular wavelengths rather than in a continuum, showing directly that atomic energies are restricted to discrete values.

The three phenomena work by different mechanisms yet point at the same thing, and the value of h extracted from each agrees. It is this cross-domain consistency, not the black-body fit alone, that made the hypothesis acceptable.

13 The interface: what is still missing before de Broglie

13.1 $\varepsilon = h\nu$ versus $E = \hbar\omega$

These are the same relation in different notation: $\varepsilon = h\nu$ becomes $\varepsilon = \hbar\omega$ on substituting $\omega = 2\pi\nu$ and $\hbar = h/2\pi$.

Their standing differs, and this must not be blurred:

- Planck’s relation is about **electromagnetic field modes**: the energy of a mode is an integer multiple of $h\nu$. It asserts discreteness of energy.

- The postulate's relation is about **any particle**: the particle's energy is proportional to the frequency of its associated wave. It asserts a correspondence between particle and wave.

The first does not entail the second. Between “light comes in portions” and “an electron has a frequency” lies the gap.

13.2 The photon segment is derivable

[IOU] $E^2 = (pc)^2 + (mc^2)^2$ comes from special relativity; this IOU is assigned to Chapter 07. For a photon $m = 0$, giving $E = pc$. Combined with $\nu = c/\lambda$ (which is just the Chapter 04 dispersion relation $\omega = ck$ rewritten: $2\pi\nu = c \cdot 2\pi/\lambda$, cancelling 2π), this converts $E = h\nu$ into $p = h/\lambda$.

Note that this IOU points forwards: Chapter 05 uses a result from Chapter 07, the only place in the list where the dependency runs backwards against the numbering.

13.3 The step that is not derivable

Carrying the relation over from light to massive particles gives $\lambda = h/p$. Mathematically this does not go through, and it was never going to: logically it is not a derivation but a new axiom.

So the correct way to write it is not “generalising from photons to particles” (which pretends to derive) but:

Postulate: every particle has an associated wave satisfying $p = \hbar k$ and $E = \hbar\omega$.

The photon segment serves only to show that the postulate is consistent with known results in the $m = 0$ case. It is a **consistency check**, not a link in the derivation. Presenting it as a step is exactly what makes the passage read awkwardly.

When de Broglie wrote his thesis in 1924 he had no direct experimental support (electron diffraction came in 1927). His actual reasons were three, none of them derivations:

1. **A belief in symmetry:** light had been taken to be purely a wave until the photoelectric effect forced particle behaviour on it; he asked the converse question, whether things taken to be particles might have wave behaviour.
2. **It explains an outstanding puzzle:** Bohr's 1913 model contains an unexplained rule, $L = n\hbar$. If the electron is a wave and the orbit must accommodate a whole number of wavelengths (a periodic boundary condition, joining head to tail, hence whole wavelengths rather than half ones), then $2\pi r = n\lambda$ with $\lambda = h/p$ gives exactly $L = n\hbar$. A rule that had been inserted by hand becomes a consequence of a standing-wave condition — his strongest argument. Its logical standing, though, is a post hoc consistency check, and Bohr's rule was itself inserted by hand.
3. **Relativistic phase consistency:** requiring the particle's internal oscillation and its phase wave to stay in step across reference frames. This part is contested today but was his point of departure.

What eventually established the postulate was electron diffraction, neutron diffraction, and the success of the whole of quantum mechanics once it had been absorbed into the Schrödinger equation.

None of this depends on the size of the mass — mass only sets how small the wavelength is (for macroscopic objects, too small to detect), not whether the assumption holds. Nor does it constitute a rift between quantum mechanics and relativity: $E^2 = (pc)^2 + (mc^2)^2$ holds for both kinds of particle, and so do $p = \hbar k$ and $E = \hbar\omega$. The Schrödinger equation cannot handle photons because it uses the non-relativistic energy $E = p^2/2m$, which is a question of that equation's range of validity, and the relativistic quantum equations of Chapter 08 exist to repair it. The genuine rift is between quantum mechanics and general relativity (Chapter 17).

[IOU] De Broglie's postulate is not derivable at the level of Chapter 05. Its provenance lies in quantum field theory (Chapter 09): there the particle is not the basic object, the field is; an electron is an excitation of the electron field, so wave behaviour is the field's nature, and $p = \hbar k$ follows from translational symmetry of the field. The price is a new set of postulates (the existence of fields, commutation relations, the form of the Lagrangian). So this is not closure but relocation of the IOU one layer down.

13.4 What became of wave–particle duality

Not an explanation, but one piece of mathematics describing both behaviours. Each particle is associated with a wave function ψ evolving by the Schrödinger equation — this is the wave part, which interferes and superposes; a measurement yields a definite position with probability $|\psi|^2$ — this is the particle part. One ψ carries both behaviours, so they are no longer two mutually contradictory models.

Mathematically the contradiction is gone. But “why a measurement yields a single outcome” is not answered by the mathematics — that is the measurement problem, with several mutually incompatible interpretations and no experiment yet able to separate them. Quantum field theory pushes the picture one step further and dissolves the question “wave or particle” altogether, but the measurement problem remains.

14 The list of IOUs

IOU	content	when repaid
Kirchhoff's law	spectrum depends only on T , not material, shape or size	not descended into
Maxwell's equations	the provenance of the 3D wave equation	not descended into
The m in the oscillator	replaced by a field constant, not a mass	not descended into
Equipartition	the integral giving $\bar{E} = k_B T$ for continuous energy	not descended into
Wien's general form	the adiabatic-compression derivation of $\rho = \nu^3 f(\nu/T)$	undecided
$E = pc$	$E^2 = (pc)^2 + (mc^2)^2$ with $m = 0$	Ch. 07, special relativity
De Broglie's postulate	why particles are waves	Ch. 09, quantum field theory

Separately: thermodynamics and statistical mechanics have no chapter of their own in the nineteen-item list, and everything used along this line (the Boltzmann distribution, equipartition, Wien's displacement law, Stefan–Boltzmann, the radiation pressure $P = u/3$) has been drawn on piece-meal. Chapter 13 (the hot universe, the CMB) will need more of it. Whether to consolidate it into its own chapter can be decided then; if so, the scope should be fixed hard by the descent criterion — only the few results this chain actually uses — since the foundations of that subject (the second law, the microscopic origin of entropy, the ergodic hypothesis) run away very easily.

A Appendix A: maximum entropy, duality and softmax

This section branches off the question “where does the Boltzmann distribution come from”. It runs parallel to the physics and is kept separate.

A.1 Maximum entropy yields the exponential family

With finitely many outcomes $y_1, \dots, y_j$, we seek a probability table $p = (p_1, \dots, p_j)$. The entropy is $H(p) = -\sum_i p_i \log p_i$. There are two groups of constraints: normalisation $\sum_i p_i = 1$, and feature expectations $\sum_i p_i f_r(y_i) = \hat{F}_r$ for $r = 1, \dots, m$, where f_r are given feature functions and $\hat{F}_r$ the empirical averages from data.

The Lagrangian:

$$\mathcal{L}(p, \theta, \lambda_0) = -\sum_i p_i \log p_i + \sum_r \theta_r \left(\hat{F}_r - \sum_i p_i f_r(y_i) \right) + (\lambda_0 - 1) \left(1 - \sum_i p_i \right)$$

Differentiating with respect to p_i and setting to zero:

$$-\log p_i - 1 - \sum_r \theta_r f_r(y_i) - (\lambda_0 - 1) = 0 \implies p_i = e^{-\lambda_0} e^{-\theta \cdot f(y_i)}$$

Eliminating λ_0 by normalisation, with $Z(\theta) = \sum_k e^{-\theta \cdot f(y_k)}$:

$$p_i(\theta) = \frac{e^{-\theta \cdot f(y_i)}}{Z(\theta)}$$

The softmax form emerges. Note the division of labour: f belongs to the data and is fixed; θ belongs to the model and is learned. The dimension of θ equals the **number of features**, and has nothing to do with the number of outcomes.

A.2 The dual function is the negative log-likelihood

Substituting $p(\theta)$ back into $\mathcal{L}$ and eliminating p and λ_0 , two terms cancel and the dual function is

$$g(\theta) = \log Z(\theta) + \sum_r \theta_r \hat{F}_r$$

while the log-likelihood of N samples is

$$\log L(\theta) = \sum_n \left[-\theta \cdot f(y^{(n)}) - \log Z(\theta) \right] = -N \left[\log Z(\theta) + \theta \cdot \hat{F} \right]$$

so $g(\theta) = -\frac{1}{N} \log L(\theta)$: minimising the dual is maximising the log-likelihood.

Differentiating g and setting to zero:

$$\frac{\partial g}{\partial \theta_r} = -\sum_i p_i f_r(y_i) + \hat{F}_r = 0$$

which is the original constraint. The loop closes.

The point to retain: maximum entropy fixes only the **form** $p \propto e^{-\theta \cdot f}$, with θ an undetermined multiplier; the constraints then pin θ to a unique value. Maximising entropy narrows the candidates from “all distributions” to the exponential family; the constraints select the one θ within it. With no constraints beyond normalisation the solution is the uniform distribution, corresponding to $\theta = 0$.

Incidentally, $\mathbb{E}_p[f_r] = -\partial(\log Z)/\partial\theta_r$ is an identity holding for any feature; the energy case is merely $f = E$, $\theta = \beta$, i.e. the $\bar{E} = -\partial(\log Z)/\partial\beta$ used in the main text. Differentiating once more gives the variance, corresponding physically to the link between energy fluctuations and specific heat, and in machine learning to $\log Z$ being the cumulant generating function.

A.3 Why duality is called duality

A minimal example: $\min x^2$ subject to $x \geq 2$ (standard form $2 - x \leq 0$).

The Lagrangian is $L(x, \lambda) = x^2 + \lambda(2 - x)$ with $\lambda \geq 0$. At any feasible point $2 - x \leq 0$ and $\lambda \geq 0$, so $L \leq x^2$ — this is where the bound comes from. Minimising over x eliminates it ($x = \lambda/2$), giving

$$g(\lambda) = 2\lambda - \frac{\lambda^2}{4}$$

The dual problem $\max_{\lambda \geq 0} g(\lambda)$ gives $\lambda = 4$ and $g(4) = 4$, equal to the primal optimum 4 (strong duality, the problem being convex), with $x = \lambda/2 = 2$ the primal solution.

The name reflects the geometry: for any $\lambda \geq 0$ and any feasible x we have $g(\lambda) \leq f(x)$, so each λ supplies a lower bound and the dual problem is the search for the tightest one. The primal approaches from above, the dual from below — a symmetric pair squeezing towards each other. In linear programming this is most visible: forming the dual transposes the constraint matrix, and the number of primal constraints equals the number of dual variables.

The multiplier is a shadow price: changing the constraint to $x \geq 2 + \epsilon$ moves the optimum from 4 to roughly $4 + 4\epsilon$, with coefficient exactly $\lambda = 4$.

The requirement $\lambda \geq 0$ for inequality constraints has the same origin: with an equality constraint $\lambda h(x)$ vanishes identically and the multiplier may take either sign; with an inequality, preserving $\lambda g(x) \leq 0$ forces a sign. Adding complementary slackness gives the KKT conditions.

Dualising a convex closed problem twice returns the original ($f^{**} = f$); without convexity f^{**} is the convex hull and one cannot get back, which is the origin of the duality gap.

A.4 Forcing the correspondence with two-tower models

In a two-tower retrieval model, $p(d|q) = \text{softmax}(\text{dis}(q, d))$. Pushing this into the framework above admits only one correspondence: arrange the k candidate documents along k dimensions, let the i -th be $\text{dis}(q, d_i)$, and set $\theta_i = 1$ (or rather, θ_i and f_i are absorbed together into $\text{dis}(q, d_i)$, with no unique way to separate them), leaving the deep model to place each pair (q, d_i) at that position.

The price: θ degenerates and has no free parameters, so the expectation constraint (the model’s expected dis equalling the average dis over clicked pairs) is hidden in a two-tower model and carries no force — it is a by-product of convergence, not the driver of learning. Two-tower models therefore **share the softmax form but not the internal $\theta \cdot f$ structure**: maximum entropy explains

why the form is exponential, and the two-tower model simply adopts that form as a design choice. What does match the framework strictly is the linear case, with f given by hand and θ learned: logistic regression, CRFs, maximum-entropy classifiers. The reading for deep models is that the final linear layer plays the role of θ and everything before it is a learned feature generator; this identity is exact only when the feature layers are frozen, and during training, with the features moving, it holds only instantaneously. In multi-class classification each class has a row of weights and the logit is the inner product of that row with the features — documents occupy the position of classes, which is where the two-tower model and the classifier coincide.

One further difference of kind: documents are finite in number and normalisation is a sum, whereas energy is continuous and the sum becomes an integral, with softmax becoming a probability density.

Finally, a softmax logit is a log-odds in the binary case but not in the multi-class case — there z_i is $\log p_i$ plus a common constant $\log Z$, and only differences $z_i - z_j = \log(p_i/p_j)$ carry meaning. Two-tower models are the latter case, so their scores can rank but cannot serve directly as probabilities; changing the candidate set changes Z and shifts the whole array of $\log p_i$ (while dis itself is unchanged).